

Lieb-Schultz-Mattis Theorem for Open Quantum Systems

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- ① Yi-Neng Zhou, Xingyu Li, Hui Zhai, Chengshu Li, and Yingfei Gu,
Reviving the Lieb-Schultz-Mattis Theorem in Open Quantum Systems, arXiv:2310.01475
- ② Kohei Kawabata, Ramanjit Sohal, and Shinsei Ryu,
Lieb-Schultz-Mattis Theorem in Open Quantum Systems,
Phys. Rev. Lett. **132**, 070402 (2024)

Theorem (Lieb-Schultz-Mattis)

A one-dimensional periodic chain of length L , with half-integer spin per unit cell, has an excitation gap bounded by $\text{const.}/L$.

So there are three possible scenarios for 1D spin systems:

- 1 Gapless spectrum
- 2 Gapped spectrum with spontaneous symmetry breaking
- 3 Gapped spectrum with ground state degeneracy

LSM theorem provides the restrictions of 1D quantum systems.

Introduction

Essence of LSM theorem

- 1 Define the twist operator

$$\hat{U}_{\text{twist}} := \exp \left(i \frac{2\pi}{L} \sum_j j \hat{S}_j \right)$$

to construct a state $|\psi'\rangle = \hat{U}_{\text{twist}} |\psi\rangle$ such that $\langle\psi|\psi'\rangle = 0$.

- 2 The eigenvalue of $|\psi'\rangle$ is either the ground state energy (ground state degeneracy) or the energy of the first excited state.
- 3 For spin-N, such a statement fails (since $[\hat{T}, \hat{U}_{\text{twist}}] = 0$), leads to the possibility for gap opening (Haldane gap).

MOTIVATION: The discussion by LSM is based on a isolated system.
Then **how about the open quantum systems?**

In open quantum systems, **energy spectrum is no longer well-defined**.
Physicists need to find a **new theoretical framework** and the **new quantity** playing the role in the open quantum systems as the spectral gap did in the original one.

Literature 1

arXiv:2310.01475

Entanglement Spectrum

Definition (Entanglement Hamiltonian)

$$\hat{\rho}_s = e^{-\hat{K}}$$

where $\hat{\rho}_s$ is the reduced density matrix of the system. $\hat{K}$ is the *entanglement Hamiltonian*, whose spectrum is called the *entanglement spectrum*.

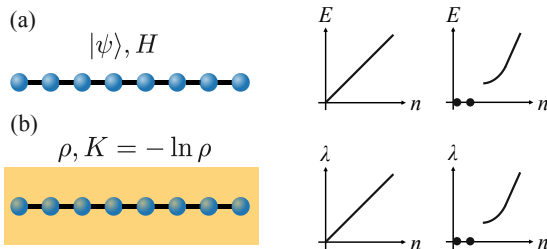


Figure: The entanglement spectrum has the similar behaviour to the energy spectrum.

Entanglement LSM Theorem

Theorem (Entanglement LSM)

The entanglement spectrum is gapless or has the ground state degeneracy if

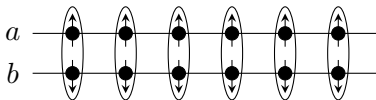
- *Half-integer representation (ex: spin S or filling factor ν)*
- *Onsite rotational symmetry is preserved.*
- *Short-range correlation behaviour: $\langle S_i^z S_j^z \rangle \sim e^{-|i-j|/\xi}$*

The **short-range correlation behaviour** is crucial in the latter discussion. Such a phenomenon is caused by the coupling to the environment.

- 1 $\xi \sim 1/\Delta$. The system gap Δ is opened by the coupling to the bath.
- 2 $\xi \sim 1/T$. Due to the thermalization.

Entanglement LSM Theorem (Theoretical Intuition)

Example: Coupling between two spin chains. Spin chain a is the system and b is the bath (environment).



$$\hat{H} = \left(J \sum_i \hat{\mathbf{S}}_{i,a} \cdot \hat{\mathbf{S}}_{i+1,a} + J \sum_i \hat{\mathbf{S}}_{i,b} \cdot \hat{\mathbf{S}}_{i+1,b} \right) + \Delta \sum_i \hat{\mathbf{S}}_{i,a} \cdot \hat{\mathbf{S}}_{i,b}$$

$$|0\rangle = \frac{1}{2^{L/2}} \bigotimes_{i=1}^L \left(|\uparrow\rangle_{i,a} |\downarrow\rangle_{i,b} - |\downarrow\rangle_{i,a} |\uparrow\rangle_{i,b} \right) = \frac{1}{2^{L/2}} \sum_n |n\rangle_a |\bar{n}\rangle_b$$

Entanglement LSM Theorem (Theoretical Intuition)

Apply the perturbation theory,

$$|\psi\rangle \approx |0\rangle - \frac{\hat{H}_a + \hat{H}_b}{2\Delta} |0\rangle \approx e^{-\frac{1}{\Delta}(\hat{H}_a + \hat{H}_b)/2} |0\rangle$$

The reduced density matrix of the system, spin chain a , is then

$$\hat{\rho}_s = \text{Tr}_b (|\psi\rangle \langle\psi|) = \frac{1}{2^L} \sum_n e^{-2\frac{1}{\Delta} E_n} |n\rangle_a \langle n|_a = \frac{1}{2^L} e^{2\frac{1}{\Delta} \hat{H}}$$

Based on this, the entanglement Hamiltonian is connected with the Hamiltonian we are familiar with,

$$\hat{K} \sim \frac{1}{\Delta} \hat{H}$$

- The entanglement spectrum satisfies the LSM theorem.

Question: An **information theoretical description** of the **locality** of the entanglement Hamiltonian.

Goal: To show that the spectral gap of $\hat{K}$ vanishes in the $L \rightarrow \infty$ limit.

► Mutual Information Function

$$I(A : B) \equiv S_A + S_B - S_{A \cup B}$$

$$\begin{aligned} I(A : C|B) &\equiv I(A : B \cup C) - I(A : B) \\ &= I(C : B \cup A) - I(C : B) \\ &= S_{A \cup B} + S_{C \cup B} - S_B - S_{A \cup B \cup C} \end{aligned}$$

The conditional mutual information $I(A : C|B)$ measures the information coupling between A and C bridged by B .

Locality and Mutual Information

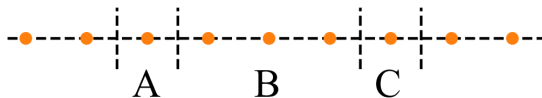


Figure: The ground state of the system discussed in the literature is a quantum approximate Markov state with $\epsilon \sim e^{-l/\xi}$, where l is the length scale of the bridge subsystem B .

- **Quantum Markov state:** $I(A : C|B) = 0$

Which implies

$$S_{AUBUC} = S_{AUB} + S_{CUB} - S_B \quad (1)$$

indicating the locality.

- **Quantum approximate Markov state:** $I(A : C|B) < \epsilon \sim e^{-l/\xi}$

Locality and Mutual Information

- ▶ According to the authors, equation (1) is equivalent to saying

$$\hat{K}_{AUBUC} = \hat{K}_{AUB} + \hat{K}_{CUB} - \hat{K}_B$$

That is to say, the **entanglement Hamiltonian is local**, namely it could be decomposed into a sum of the entanglement Hamiltonian for smaller patches.

- ▶ For quantum approximate Markov states, the entanglement entropy is exponentially local characterized by $e^{-l/\xi}$
- ▶ However, **the statement that the state here is a quantum approximate Markov state has not yet been rigorously proved.** So such a statement is just based on the numerical result.

Numerical Result

$$\hat{H} = J_1 \sum_{i=1}^L \left(\hat{\mathbf{S}}_i \cdot \hat{\mathbf{S}}_{i+1} + \frac{1}{3} \left(\hat{\mathbf{S}}_i \cdot \hat{\mathbf{S}}_{i+1} \right)^2 \right) + J_2 \sum_{i=1}^L \hat{\mathbf{S}}_{i,s} \cdot \hat{\mathbf{S}}_{i,b}$$

where $\hat{\mathbf{S}}_i = \hat{\mathbf{S}}_{i,s} + \hat{\mathbf{S}}_{i,b}$. The system s is a spin-1/2 chain and the bath b is a spin-3/2 chain.

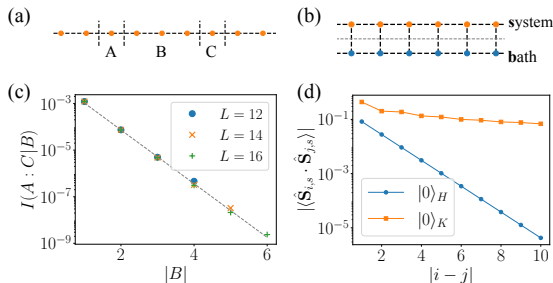


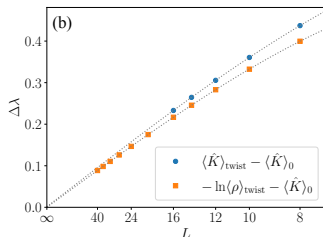
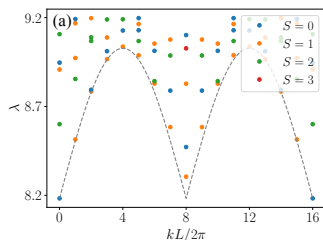
Figure: (c) Shows the behaviour of $I(A : C|B) \sim e^{-l/\xi}$.

The $e^{-l/\xi}$ behaviour leads to the vanishing of the entanglement spectral gap,

$$\begin{aligned}\langle \hat{K} \rangle_{\text{twist}} - \langle \hat{K} \rangle_0 &\sim L \int_0^\infty e^{-l/\xi} \left(1 - \cos \left(\frac{2\pi l}{L} \right) \right) dl \\ &\sim \frac{4\pi^2 \xi^3}{L} - \frac{16\pi^4 \xi^5}{L^3} + \dots\end{aligned}$$

This once again shows the importance of the locality $e^{-l/\xi}$. If the entanglement Hamiltonian is not local, $\xi \rightarrow \infty$, then the upper bound of the entanglement spectral gap is broken down.

Gapless Spectrum



- The upper bounds of the entanglement spectral gap:

$$\begin{aligned}\Delta\lambda &\leq -\ln\langle\hat{\rho}\rangle_{\text{twist}} - \langle\hat{K}\rangle_0 \\ &\leq \langle\hat{K}\rangle_{\text{twist}} - \langle\hat{K}\rangle_0\end{aligned}$$

- Curve fitting:

$$\Delta\lambda = c_0 + c_1 L^{-1} + c_3 L^{-3}$$

Ground State Degeneracy

Model: Spin-1/2 Majumdar-Ghosh (MG) chain coupled to a spin-3/2 bath

$$\hat{H} = J_1 \sum_{i=1}^L \left(\hat{\mathbf{S}}_{i,s} \cdot \hat{\mathbf{S}}_{i+1,s} + \frac{1}{2} \hat{\mathbf{S}}_{i,s} \cdot \hat{\mathbf{S}}_{i+2,s} \right) + J_2 \sum_{i=1}^L \hat{\mathbf{S}}_{i,s} \cdot \hat{\mathbf{S}}_{i,b} + D \sum_{i=1}^L \left(\hat{S}_{i,s}^z + \hat{S}_{i,b}^z \right)^2$$

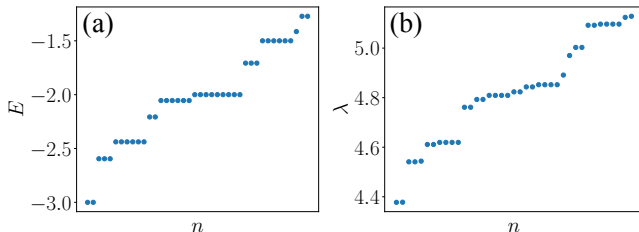


Figure: (a) The energy spectrum of a closed system (b) The entanglement spectrum. The two-fold ground state degeneracy is observed.

Questions

Here I list my questions to this paper

- Is the entanglement spectrum measurable?
- The rigorous proof of the exponential decay behaviour of the mutual information, i.e. the quantum approximate Markov state.

Literature 2

Phys. Rev. Lett. **132**, 070402 (2024)

Lindblad Equation and Dissipative Gap

Motivation: Open quantum systems are no longer described by Hamiltonians, but instead by **Liouvillians**.

The master equation describing the evolution of the density matrix is the **Lindblad equation**

$$\frac{d}{dt}\hat{\rho}_s = \mathcal{L}\hat{\rho}_s = -i[\hat{H}, \rho_s] + \sum_n \left[\hat{L}_n \hat{\rho} \hat{L}_n^\dagger - \frac{1}{2} \{ \hat{L}_n^\dagger \hat{L}_n, \hat{\rho} \} \right]$$

$\mathcal{L}$ is called the *Lindbladian*

Question: If we can find a quantity in a open quantum system playing the similar role as the spectral gap in the LSM theorem?

Lindblad Equation and Dissipative Gap

The answer given by the authors is the **dissipative gap** Δ , which is the negative real part of the eigenvalue of the Lindbladian of the first decaying state.

A simple comprehension: From the lindblad equation, we have

$$\hat{\rho}_s(t) = e^{t\mathcal{L}}\hat{\rho}_s(0)$$

If $\hat{\rho}_s(0)$ is an eigenvector of $\mathcal{L}$ with eigenvalue $iE - \Delta$, then it leads to

$$\hat{\rho}_s \sim e^{-t\Delta}$$

indicating the dissipating behaviour (Δ^{-1} is the relaxation time scale).

References

- *Phys. Rev. Lett.* **111**, 150403 (2013)
- *Phys. Rev. E* **92**, 042143 (2015)
- *Phys. Rev. A* **107**, 033332 (2023)
- *The Theory of Open Quantum Systems*, Oxford University Press

LSM Theorem in Open Quantum Systems

Mapping: $\hat{\rho} = \sum_{ij} \rho_{ij} |i\rangle \langle j| \mapsto |\rho\rangle = \sum_{ij} \rho_{ij} |i\rangle_+ |j\rangle_- \in \mathcal{H} = \mathcal{H}_+ \otimes \mathcal{H}_-$

$$\mathcal{L} = -i\hat{H}_+ + i\hat{H}_- + \sum_n \left[\hat{L}_{n,+} \hat{L}_{n,-}^* - \frac{1}{2} \left(L_{n,+}^\dagger \hat{L}_{n,+} + \hat{L}_{n,-}^T L_{n,-}^* \right) \right]$$

so now the Lindbladian becomes $\mathcal{L} : \mathcal{H} \rightarrow \mathcal{H}$

Goal: Constraint the steady state ($\mathcal{L}\hat{\rho} = 0$) and the dissipative gap solely by **symmetry**.

- Lattice translational symmetry $\mathcal{T}$
- U(1) symmetry

The conserved quantity corresponding to the U(1) symmetry is the $N_\pm$
The **filling number** is defined by $\nu := N_\pm/V$, where V is the system size
(here we focus on the subspace with $N_+ = N_-$)

LSM Theorem in Open Quantum Systems

Theorem

*In open quantum systems with **lattice translation invariance** and **strong $U(1)$ symmetry**, if the Lindbladian is gapped and exhibits a unique steady state in the subspace with the fixed $U(1)$ charge, the filling number ν is required to be an integer.*

In other words, if ν is not an integer, the Lindbladian is gapless or exhibits degenerate steady states.

LSM Theorem in Open Quantum Systems

Twist Operator: $U_{\pm}^{(\text{twist})} := e^{2\pi i \sum_{j=1}^L j n_{j,\pm}/L}$ (which is equivalent to $U(1)$ flux insertion)

$$\mathcal{T} U_+^{(\text{twist})} \mathcal{T}^{-1} = e^{-2\pi i N_+/L} U_+^{(\text{twist})} = e^{-2\pi i \nu} U_+^{(\text{twist})}$$

Spin version: $U_{\pm}^{(\text{twist})} := e^{2\pi i \sum_{j=1}^L j S_{j,\pm}^z/L}$

$$\mathcal{T} U_+^{(\text{twist})} \mathcal{T}^{-1} = e^{2\pi i S_1^z - 2\pi i S_+/L} U_+^{(\text{twist})} = e^{-2\pi i (S + S_+/L)} U_+^{(\text{twist})}$$

So the filling number for spin systems is defined by $\nu = S + S_{\pm}/L$

LSM Theorem in Open Quantum Systems

► $[\mathcal{T}, U_{\pm}^{(\text{twist})}] = 0$ only when $\nu \in \mathbb{Z}$. Otherwise, one can use $U_{\pm}^{(\text{twist})}$ to construct another state $|\psi'\rangle = U_{\pm}^{(\text{twist})} |\psi\rangle$ that is orthogonal to the original $|\psi\rangle$.

Consider a steady state $|\rho_0\rangle$. If $\nu \notin \mathbb{Z}$, there exists another steady state $|\rho'_0\rangle = U_{+}^{(\text{twist})} |\rho_0\rangle$? However, since $U_{\pm}^{(\text{twist})}$ is equivalent to $U(1)$ gauge flux insertion (twist the boundary condition), the Lindblad gap shouldn't be opened or closed by such an operation.

Conclusion: The gapped Lindbladian with the unique steady state requires the integer filling number.

Numerical Result

$$\hat{H} = \sum_{n=1}^L [J (S_n^x S_{n+1}^x + S_n^y S_{n+1}^y) + J_z S_n^z S_{n+1}^z]$$

$$\hat{L}_n = \sqrt{\gamma} S_n^z$$

This system respects U(1) symmetry ($U_{\pm} = e^{i\theta S_{\pm}^z}$), $[\mathcal{L}, S_{\pm}^z] = 0$. Thus the conserved quantity is the total z -magnetization $S^z := \sum_{n=1}^L S_{n,\pm}^z$, and the filling number is $\nu = S_{\pm}^z / L + S$.

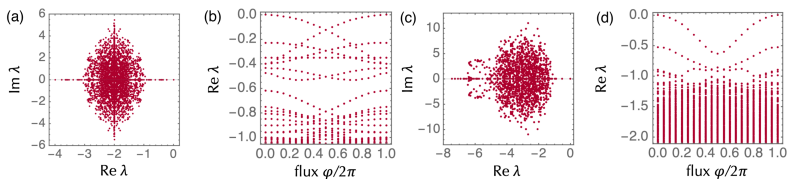


Figure: Dissipative Heisenberg XXZ model with $J = J_z = \gamma = 1.0$. (a)(b) spin-1/2, $L = 8$, (c)(d) spin-1, $L = 5$. Here the authors set $S_{\pm}^z = 0$ (no net magnetization)

Symmetry Anomaly

Conclusion

- The energy spectrum is not well defined in **open quantum systems**, and the system is better described by the Lindblad equation instead of the Schrödinger equation for closed systems.
- The LSM theorem is still applicable in open quantum systems but the spectral gap should be replaced by the **entanglement spectral gap** or the **dissipative gap**.
- Other phenomena related to LSM theorem, e.g. the symmetry anomaly, can also be discussed in these theoretical frameworks.

THANKS FOR LISTENING

Gauge transformation and twist operator

References:

- *Phys. Rev. Lett.* **78**, 1984 (1997)
- *Phys. Rev. Lett.* **84**, 1535 (1999)